

A DENSITY-ONE LOWER BOUND FOR THE FIRST DECREASE OF A HARMONIC DENOMINATOR

CHATGPT

ABSTRACT. For a positive integer a , let $b(a)$ be the smallest integer $b > a$ such that the denominator of

$$\frac{1}{a} + \frac{1}{a+1} + \cdots + \frac{1}{b}$$

is smaller than the denominator of the same sum ending at $b-1$. We prove that

$$b(a) > a + \exp\left(\frac{1}{2}\sqrt{\log a \log \log a}\right)$$

for almost all positive integers a . In particular, for every fixed $C > 0$ we have $b(a) > a + (\log a)^C$ for almost all a .

1. INTRODUCTION

For positive integers $a < b$, let $u_{a,b}$ and $v_{a,b}$ be the coprime positive integers for which

$$\sum_{i=a}^b \frac{1}{i} = \frac{u_{a,b}}{v_{a,b}}.$$

All logarithms in this paper are natural. For a given $a \in \mathbb{N}$, let $b(a)$ be the smallest integer $b > a$ for which $v_{a,b} < v_{a,b-1}$. The existence of $b(a)$ for every a was asked by Erdős and Graham in [3] and proved in [1]. More recently, the exact positive value of

$$\liminf_{a \rightarrow \infty} \frac{b(a) - a}{\log a}$$

was determined in [2]. Thus there are infinitely many a for which $b(a) - a$ has logarithmic order. The result below shows that these values of a form a set of density zero.

Theorem 1. *For almost all positive integers a we have*

$$b(a) > a + \exp\left(\frac{1}{2}\sqrt{\log a \log \log a}\right).$$

Equivalently, as $x \rightarrow \infty$,

$$\#\left\{a \leq x : b(a) \leq a + \exp\left(\frac{1}{2}\sqrt{\log a \log \log a}\right)\right\} = o(x).$$

Corollary 2. *For every fixed $C > 0$, we have*

$$b(a) > a + (\log a)^C$$

for almost all positive integers a . In particular, there are infinitely many positive integers a for which $b(a) > a + (\log a)^2$.

2. A NECESSARY CONDITION FOR A DECREASE

For a positive integer n , let

$$L_n := \text{lcm}(1, 2, \dots, n).$$

We start with a necessary condition for the denominator to decrease.

Lemma 3. *For all positive integers $a < b$, we have*

$$\frac{v_{a,b}}{v_{a,b-1}} \geq \frac{b}{\gcd(b, L_{b-a})^2}.$$

Consequently, if $v_{a,b} < v_{a,b-1}$, then

$$\gcd(b, L_{b-a}) > \sqrt{b}.$$

Proof. Write

$$\sum_{i=a}^{b-1} \frac{1}{i} = \frac{U}{V},$$

where U and V are coprime positive integers, so that $V = v_{a,b-1}$. Let $q = \gcd(V, b)$ and write $V = qV'$ and $b = qb'$. We then have $\gcd(V', b') = 1$ and

$$\frac{U}{V} + \frac{1}{b} = \frac{Ub' + V'}{qV'b'}.$$

Since $\gcd(U, V') = \gcd(V', b') = 1$, the numerator $Ub' + V'$ is coprime to both V' and b' . Hence the only possible cancellation is with q , and

$$\frac{v_{a,b}}{v_{a,b-1}} = \frac{b'}{\gcd(Ub' + V', q)} \geq \frac{b'}{q} = \frac{b}{q^2}.$$

Since V divides $\text{lcm}(a, a+1, \dots, b-1)$, we have

$$q \leq \gcd(b, \text{lcm}(a, a+1, \dots, b-1)).$$

For every prime power $p^k \mid b$, there is a multiple of p^k in the interval $[a, b-1]$ if and only if $b - p^k \geq a$, or equivalently $p^k \leq b - a$. It follows that

$$\gcd(b, \text{lcm}(a, a+1, \dots, b-1)) = \gcd(b, L_{b-a}).$$

This proves the first claim. If $v_{a,b} < v_{a,b-1}$, the first claim gives $b/\gcd(b, L_{b-a})^2 < 1$, which proves the second claim. $\square$

3. PROOF OF THEOREM 1

Proof. Let X be large and set

$$N := \left\lfloor \exp \left(\frac{1}{2} \sqrt{\log(2X) \log \log(2X)} \right) \right\rfloor.$$

We count the integers $a \in [X, 2X)$ for which $v_{a,a+n} < v_{a,a+n-1}$ for some $1 \leq n \leq N$. By Lemma 3, every such pair (a, n) satisfies

$$\gcd(a+n, L_n) > \sqrt{a+n} > \sqrt{X}.$$

Moreover, this greatest common divisor divides L_N and $a + n$. Since $N < X$ for all sufficiently large X , it is also smaller than $3X$. Therefore the number of integers under consideration is at most

$$(1) \quad N \sum_{\substack{d|L_N \\ \sqrt{X} < d < 3X}} \left(\frac{X}{d} + 1 \right).$$

Let $0 < \delta \leq \frac{1}{2}$ and define

$$E := \prod_{p \leq N} (1 - p^{-1+\delta})^{-1}.$$

Since every divisor of L_N only has prime factors at most N , we have

$$\sum_{d|L_N} d^{-1+\delta} \leq E.$$

Moreover, $d < 3X$ implies $X/d + 1 < 4X/d$. Hence, after dividing (1) by X , the proportion of integers $a \in [X, 2X)$ under consideration is

$$(2) \quad \ll N \sum_{\substack{d|L_N \\ d > \sqrt{X}}} \frac{1}{d} \leq NEX^{-\delta/2}.$$

We next estimate E . Since $\delta \leq \frac{1}{2}$, we have

$$\log E \ll \sum_{p \leq N} p^{-1+\delta}.$$

By partial summation and the prime number theorem,

$$\sum_{p \leq N} p^{-1+\delta} \ll \frac{N^\delta}{\log N} + \int_2^N \frac{t^{-1+\delta}}{\log t} dt.$$

If $\delta \log N \geq 1$, splitting the integral at $e^{1/\delta}$ gives

$$(3) \quad \log E \ll \log \left(\frac{2}{\delta} \right) + \frac{N^\delta}{\delta \log N}.$$

Indeed, the part below $e^{1/\delta}$ is $\ll \log(2/\delta)$. After the substitution $u = \delta \log t$, the part above $e^{1/\delta}$ is bounded by a constant multiple of

$$\int_1^{\delta \log N} \frac{e^u}{u} du \ll \frac{N^\delta}{\delta \log N}.$$

Put $y = \log X$ and choose

$$\delta := \frac{\frac{1}{2} \log y + \frac{1}{4} \log \log y}{\log N}.$$

For sufficiently large X , this choice satisfies $0 < \delta \leq \frac{1}{2}$ and $\delta \log N \geq 1$. Moreover,

$$N^\delta = y^{1/2} (\log y)^{1/4},$$

so (3) gives

$$(4) \quad \log E \ll \log y + y^{1/2} (\log y)^{-3/4}.$$

On the other hand, since

$$\log N = \frac{1}{2}\sqrt{y \log y} + o(1),$$

we have

$$\begin{aligned} \frac{\delta y}{2} - \log N &= \frac{y}{2 \log N} \left(\frac{1}{2} \log y + \frac{1}{4} \log \log y \right) - \log N \\ (5) \qquad &= \left(\frac{1}{4} + o(1) \right) \sqrt{\frac{y}{\log y}} \log \log y. \end{aligned}$$

The right-hand side of (4) is

$$o \left(\sqrt{\frac{y}{\log y}} \log \log y \right).$$

Combining this with (5), we obtain

$$\log \left(N E X^{-\delta/2} \right) = \log N + \log E - \frac{\delta y}{2} \longrightarrow -\infty.$$

By (2), only $o(X)$ integers $a \in [X, 2X)$ admit a decrease within the next N terms. The function

$$\exp \left(\frac{1}{2} \sqrt{\log x \log \log x} \right)$$

is increasing for all sufficiently large x . Hence every $a \in [X, 2X)$ for which

$$b(a) \leq a + \exp \left(\frac{1}{2} \sqrt{\log a \log \log a} \right)$$

is among the integers just counted. Thus the number of exceptions in $[X, 2X)$ is $o(X)$. A standard dyadic decomposition now shows that the number of exceptions up to x is $o(x)$, proving Theorem 1. $\square$

Proof of Corollary 2. For every fixed $C > 0$, we have

$$\frac{1}{2} \sqrt{\log a \log \log a} > C \log \log a$$

for all sufficiently large a . The corollary follows from Theorem 1. $\square$

REFERENCES

- [1] W. van Doorn, *On the non-monotonicity of the denominator of generalized harmonic sums*, arXiv:2411.03073 (2024).
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